# SOURCE-RESOLVED ERROR CALCULUS FOR PROJECTED MULTILINEAR COMPUTATIONAL GRAPHS

ELIUTH CHAVERO JASSO

Abstract. Recursive projection creates local closure residuals that can be propagated through trees and directed acyclic graphs. This paper consolidates a finite source-resolved calculus developed in the repository: scalar DAG majorants, first-order vector provenance, exact higher-order source polynomials with repeated-source multi-indices, and signed compositional aggregation. The central refinement is to combine coefficients by source and multi-index before taking norms, yielding certified inequalities $B_{\text{actual}} \le B_{\text{signed}} \le B_{\text{treewise}}$ and strict cancellation witnesses. Associator, Jacobiator, and Filippov defects are treated as instances of one signed-expression engine; no identity is assumed to vanish. Approximate-law representation error and validated norm enclosures are recorded as separate certificate layers.

## 1. Why a graph calculus?

Tree certificates are safe but can overcount shared subcomputations. If one source fans out and returns through multiple paths, a pathwise triangle bound norms every route before remembering that the routes originated from the same error. Signed identities introduce a second loss: independently bounding each term destroys cancellation between bracketings or forest terms. The v4/v5 implementation addresses both losses while preserving the finite algebraic semantics of recursive projection.

## 2. Typed DAG model

Let $G = (V, E)$ be a finite typed DAG with a designated output node. Each node $v$ has a multilinear law $\mu_v$ and receives already computed values from its predecessors. Let $F_v$ denote the ambient value and $R_v$ the recursively projected value. The local difference is $D_v = F_v - R_v$. A local normal source is labelled by $s$ and written $\varepsilon_s$; it may be a closure residual, an approximate-law residual, or another explicitly registered source family.

For scalar nonnegative summaries, assume

$$B_v \le \lambda_v + \sum_{u \to v} h_{v,u} B_u, \qquad \lambda_v, h_{v,u} \ge 0. \tag{1}$$

**Theorem 2.1** (Scalar DAG source certificate)**.** *Let $\widehat{B}_v$ be defined by equality in (1) in topological order. Define $w_r = 1$ at the output $r$ and, in reverse topological order,*

$$w_u = \sum_{v : u \to v} h_{v,u} w_v.$$

*Then*

$$B_r \le \widehat{B}_r = \sum_{u \in V} \lambda_u w_u.$$

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Research manuscript. The results are finite and conditional on the stated algebraic and norm hypotheses. Novelty and independent human review remain pending.

*The calculation is $O(|V|+|E|)$ and does not enumerate paths.*

The term $\lambda_u w_u$ is a source-resolved attribution. If the output projection annihilates the root local residual, the root source is omitted from the observable projected certificate.

## 3. First-order source provenance

At first order, the error at each node can be represented as

$$D_v^{(1)} = \sum_s A_{v,s}\varepsilon_s. \tag{2}$$

The operators $A_{v,s}$ are propagated in topological order. If a source reaches the output over paths $\pi$, the source-aware operator is

$$A_{r,s} = \sum_{\pi : s \rightsquigarrow r} A_\pi.$$

Thus

$$\|D_r^{(1)}\| \le \sum_s \|A_{r,s}\|\,\|\varepsilon_s\| \le \sum_s \sum_{\pi : s \rightsquigarrow r} \|A_\pi\|\,\|\varepsilon_s\|.$$

**Corollary 3.1** (Source refinement)**.** *The first-order source-aware certificate is never larger than the pathwise triangle certificate built from the same path operators. Strict improvement is possible when routes from a shared source cancel.*

## 4. Exact nonlinear source polynomials

Sets of sources are insufficient when the same source is used in two slots. Let $\alpha = (\alpha_s)_{s\in\mathcal{S}}$ be a finite multi-index with total order $|\alpha| = \sum_s \alpha_s$. Write $[\varepsilon^\alpha]$ for the corresponding multilinear source monomial, including repeated occurrences.

**Definition 4.1** (Source polynomial)**.** The source polynomial at node $v$ is the finite formal expansion

$$\mathcal{P}_v(\varepsilon) = \sum_{\alpha\neq 0} A_{v,\alpha}[\varepsilon^\alpha], \qquad D_v = \mathcal{P}_v(\varepsilon).$$

**Theorem 4.2** (Exact finite DAG provenance)**.** *For a finite DAG of multilinear laws with finitely many labelled local sources, the source polynomial at every node is obtained exactly by topological recurrence. When a law receives source polynomials with multi-indices $\alpha$ and $\beta$ in different slots, the product term carries index $\alpha+\beta$. Consequently a repeated source used in two slots produces a term with $\alpha_s = 2$. On a tree, the construction reduces to the ordinary child-error subset expansion after forgetting global source labels.*

The exact implementation is finite for finite declared DAGs. Truncation is allowed only with an explicit tail envelope:

$$D_v = D_v^{(\le p)} + R_v^{(>p)}, \qquad \|R_v^{(>p)}\| \le \mathcal{R}_v^{(p)}.$$

The repository tests $p = 1, 2, 3$ and repeated-source, mixed-source, shared subexpression, projected-root, real, and complex controls.

## 5. Signed compositional expressions

Let

$$\mathfrak{F} = \sum_{j=1}^m c_j T_j$$

be a finite signed expression, where each $T_j$ has source coefficients $A_{j,\alpha}$. Aggregate before taking norms:

$$A_{\mathfrak{F},\alpha} = \sum_{j=1}^{m} c_j A_{j,\alpha}.$$

For source amplitudes $t_s$, define

$$B_{\text{signed}} = \sum_{\alpha} \|A_{\mathfrak{F},\alpha}\| \prod_{s} |t_s|^{\alpha_s},$$

whereas treewise aggregation gives

$$B_{\text{treewise}} = \sum_{j} |c_j| \sum_{\alpha} \|A_{j,\alpha}\| \prod_{s} |t_s|^{\alpha_s}.$$

**Theorem 5.1** (Signed source-polynomial refinement)**.** *For every finite exact source polynomial and signed expression,*

$$\boxed{B_{\text{actual}} \leq B_{\text{signed}} \leq B_{\text{treewise}}}.$$

*The second inequality follows coefficient-wise from the triangle inequality. It can be strict, including at orders $|\alpha| \geq 2$ and for exact cancellation.*

5.1. **Associator, Jacobiator, and Filippov defects.** The associator is the signed expression $T_L - T_R$. The Jacobiator and Filippov object are not assumed to vanish; the engine returns certificates for the calculated defects under the registered conventions. This distinction is essential: a small certified defect is not a proof that an arbitrary law satisfies an identity.

For a truncation order $p$, the signed certificate is paired with the combined remainder:

$$B^{(p)}_{\text{total}} = B^{(\leq p)}_{\text{signed}} + B^{(>p)}_{\text{remainder}}.$$

Omitted terms are never cancelled by expectation.

## 6. Approximate laws and norm enclosures

If $\widetilde{\mu}_v$ approximates $\mu_v$, define $\delta_v = \|\mu_v - \widetilde{\mu}_v\|_{\text{op}}$. Closure and representation sources remain separately labelled. The output budget decomposes into closure, representation, and interaction contributions, with nonlinear envelopes added only when their Lipschitz hypotheses are explicit.

Norm estimation is likewise layered. An exact formula is preferred, followed by validated interval enclosure, flattening/spectral bounds, CP-structure bounds, and Frobenius fallback. Every output records lower bound, upper bound, gap, method, and certification flag. An optimizer value alone is not called an operator norm certificate.

## 7. Status and open theorem problems

The finite scalar DAG, first-order source, exact higher-order source polynomial, signed expression, associator/Jacobiator/Filippov instantiations, validated finite norm enclosures, and approximate-law budget interfaces are implemented and tested in the v4/v5 research tracks. They are conditional on their declared finite hypotheses. Remaining theorem-level questions include global fixed-$\eta$ sharpness under repeated laws, dimension/rank reduction, topology-aware constants, globally tight multilinear spectral norms, and theorem-to-theorem novelty. The source package is therefore a closed finite calculus, not a claim of universal optimality or publication approval.

## 8. Conclusion

The resulting framework tracks not only how much error reaches an output, but which local sources generated it, how shared sources are reused, which higher orders interact, and where signed cancellation survives. This source-resolved calculus is the natural DAG extension of recursively projected multilinear tree analysis.

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Independent Researcher, Apizaco, Tlaxcala, Mexico

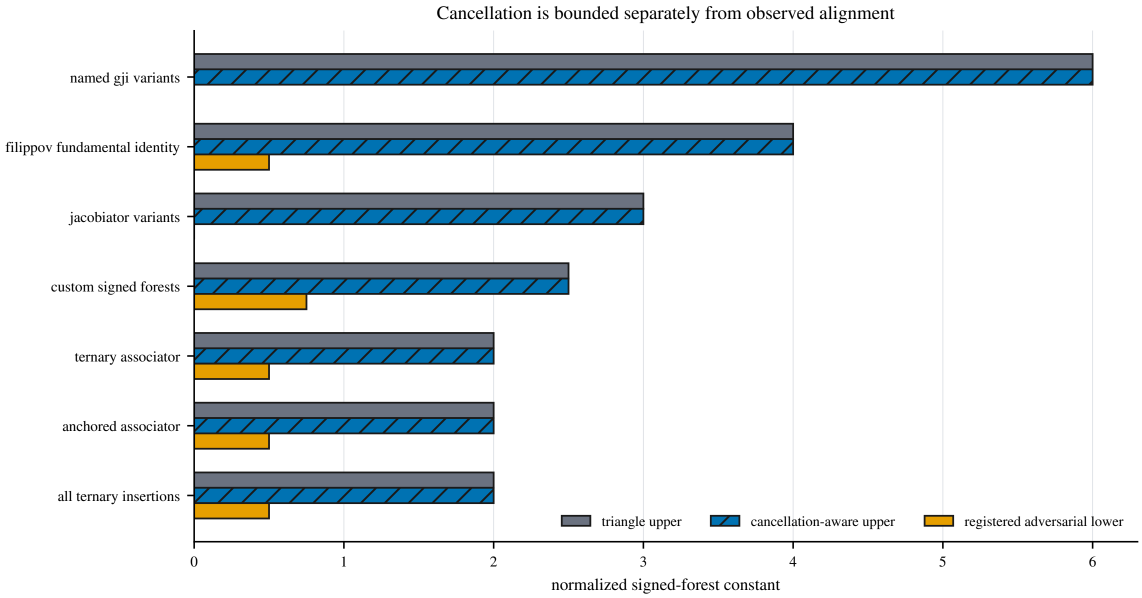


FIGURE 1. Signed source aggregation preserves cancellation before norm evaluation. Reused from the registered finite evidence package.

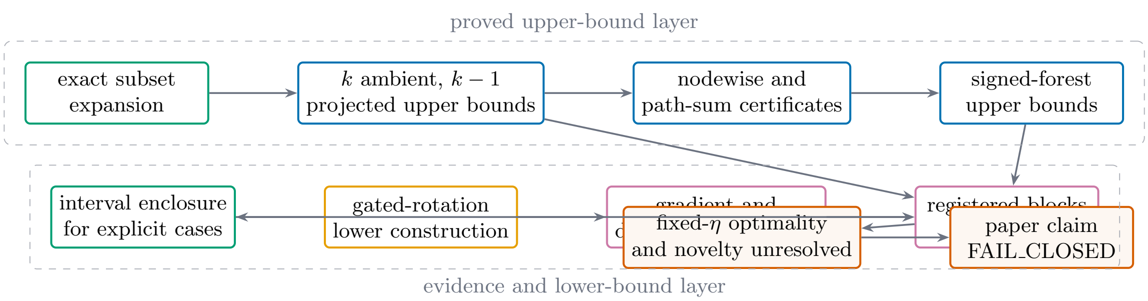


FIGURE 2. Proof, certificate, construction, empirical search, and release authorization remain separate evidence layers.